

(2) [3+3+3+3=12 points]

Carefully consider the following applications, and indicate which ADT is most appropriate for each (map, set, list, stack, or queue). Make sure to explain your reasoning. More detail is better: instead of just “List” you could state “a list of student records”.

(a) Data structure allowing you to find a book’s title from its ISBN identifier (13 characters, mostly digits, but could contain the letter ‘X’).

(b) Waitlist of students who want to enroll in CS104, but couldn’t get in yet.

(c) Data structure to store the content of each line of code of a possibly long C++ program.

(d) A data structure that allows you to input an academic semester and find all students who earned an A in CS104 that semester.